
Lecture 2: Revealed Preference

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1. From Preferences to Choices — and Back

In Lecture 1 we started from *unobservable* primitives (preferences, utility) and derived *observable* objects (demand functions). Revealed-preference theory inverts this logic:

- **Start with the observable:** the bundles a consumer actually chooses at different prices and incomes.
- **End with the unobservable:** conditions on preferences that the data are consistent with.

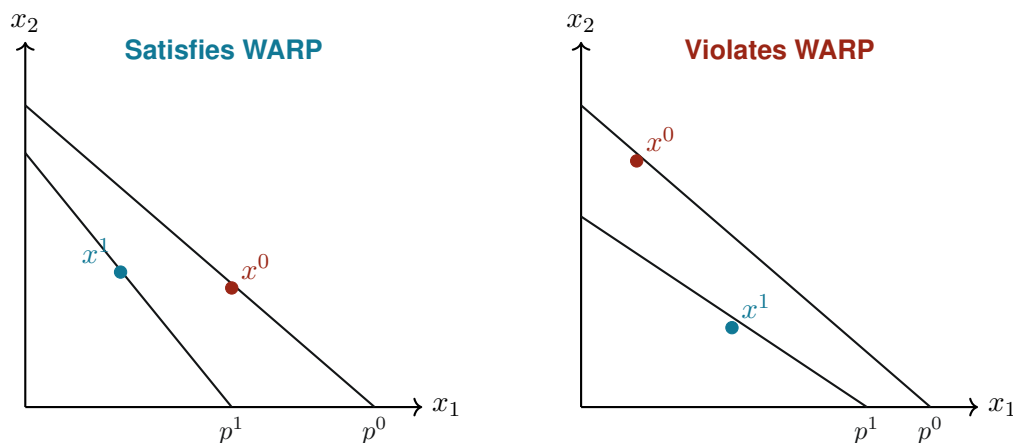
Core idea. If a consumer chooses bundle x^0 at prices p^0 , and at those prices another bundle x^1 was also affordable, then she has **revealed** that x^0 is at least as good as x^1 .

2. The Weak Axiom of Revealed Preference (WARP)

The consumer's choice behavior satisfies WARP if, for every pair of bundles x^0 and x^1 with x^0 chosen at prices p^0 and x^1 chosen at prices p^1 :

$$p^0 \cdot x^1 \leq p^0 \cdot x^0 \implies p^1 \cdot x^0 > p^1 \cdot x^1.$$

Interpretation. If x^0 is revealed preferred to x^1 (i.e. x^1 was affordable when x^0 was chosen), then x^1 can **never** be revealed preferred to x^0 . Choices must be consistent across price regimes.



In the left panel x^1 lies *strictly inside* p^0 's budget triangle, but x^0 lies *outside* p^1 's — so x^0 is revealed preferred to x^1 and the reverse cannot hold. In the right panel both bundles lie inside each other's budget set, which is a direct WARP violation.

3. The Choice Function

A **choice function** is a mapping

$$C : (p, y) \longrightarrow x \in X, \quad p \in \mathbb{R}_{++}^n, y > 0,$$

returning the bundle the consumer picks when facing prices p and income y , with the requirement that $C(p, y)$ satisfies WARP.

A choice function is **not** the same as a Marshallian demand: we never assumed it comes from utility maximization. The whole point of revealed preference is to derive demand-like properties *directly from observed choices*.

Standing assumptions.

1. Prices are strictly positive ($p \gg 0$, no free goods), so $C(p, y)$ exists.
2. **Budget balance:** $p \cdot C(p, y) = y$ (the consumer spends all her income).

4. Implications of WARP

4.1. Homogeneity of Degree Zero

If $C(p, y)$ satisfies WARP and budget balance, then for every $t > 0$:

$$C(tp, ty) = C(p, y).$$

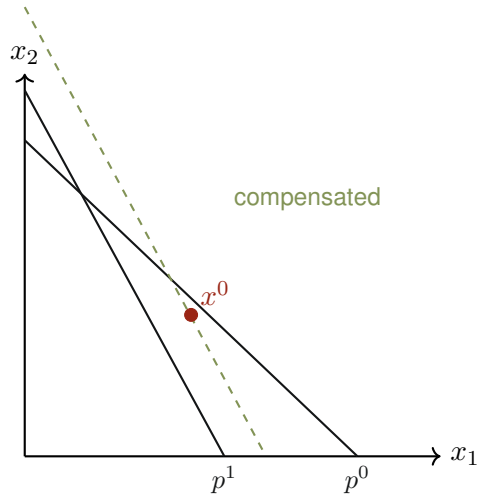
Proof. Let $x^0 = C(p^0, y^0)$ and $x^1 = C(p^1, y^1)$ with $p^1 = tp^0$ and $y^1 = ty^0$, $t > 0$. By budget balance,

$$p^1 \cdot x^1 = y^1 = ty^0 = t(p^0 \cdot x^0) = p^1 \cdot x^0,$$

so $p^1 \cdot x^1 \leq p^1 \cdot x^0$. By WARP, if $x^0 \neq x^1$ then $p^0 \cdot x^1 < p^0 \cdot x^0$. But the same equality also gives $p^0 \cdot x^1 = p^0 \cdot x^0$ (divide by t), a contradiction. Hence $x^0 = x^1$, i.e. $C(p^0, y^0) = C(tp^0, ty^0)$. ■

4.2. Income Compensation

When prices change from p^0 to p^1 , we can ask: how much income should we give the consumer so that her original choice x^0 remains *just* affordable?



Formally, if $x^0 = C(p^0, y)$ then the compensated income at p^1 is $y^1 = p^1 \cdot x^0$, and the compensated choice is

$$x^1 = C(p^1, p^1 \cdot x^0).$$

4.3. The Slutsky-Like Inequality from WARP

By construction x^0 is still affordable at (p^1, y^1) : $p^1 \cdot x^0 = y^1$. WARP then implies

$$p^0 \cdot x^0 \leq p^0 \cdot x^1, \quad (1)$$

with strict inequality unless $x^0 = x^1$. Combining (1) with the budget-balance identity $y^1 = p^1 \cdot x^1 = p^1 \cdot x^0$ and rearranging,

$$(p^1 - p^0) x^0 \geq (p^1 - p^0) C(p^1, p^1 \cdot x^0).$$

A directional formulation. Set $p^1 = p^0 + t z$ with $t \geq 0$ and $z \in \mathbb{R}^n$ an arbitrary direction. Dividing by t and taking $t \downarrow 0$ requires the auxiliary function

$$f(t) \equiv z \cdot C(p^0 + tz, (p^0 + tz) \cdot x^0)$$

to attain its maximum at $t = 0$. Hence $f'(0) \leq 0$.

4.4. The Slutsky Matrix from Choice Data

Differentiating f along the compensation path and evaluating at $t = 0$,

$$f'(0) = \sum_i \sum_j z_i \left(\frac{\partial C_i(p^0, y^0)}{\partial p_j} + x_j^0 \frac{\partial C_i(p^0, y^0)}{\partial y} \right) z_j \leq 0.$$

Since z is arbitrary, the matrix

$$S(p, y) \equiv \left[\frac{\partial C_i(p, y)}{\partial p_j} + x_j(p, y) \frac{\partial C_i(p, y)}{\partial y} \right]_{i,j}$$

is **negative semidefinite** (NSD).

This is the same Slutsky matrix from Lecture 1 — but now derived purely from choice behavior, without invoking utility maximization. The one property we *cannot* get from WARP alone is **symmetry** of S ; for that we need SARP (below).

5. Relation to Utility Maximization

5.1. Does Utility Maximization imply WARP?

If u is strictly monotone and strictly quasiconcave, the Marshallian demand

$$x^*(p, y) = \arg \max_{x \in X} u(x) \quad \text{s.t.} \quad p \cdot x \leq y$$

satisfies WARP.

Proof. Suppose $x^0 = x^*(p^0, y^0)$, $x^1 = x^*(p^1, y^1)$, $x^0 \neq x^1$, and $p^0 \cdot x^1 \leq p^0 \cdot x^0$. Then x^1 was feasible at (p^0, y^0) , so $u(x^0) \geq u(x^1)$; strict quasiconcavity and uniqueness of the optimum give $u(x^0) > u(x^1)$. Hence at prices p^1 the consumer would not choose x^0 if it were affordable — so $p^1 \cdot x^0 > p^1 \cdot x^1$. ■

5.2. Does WARP imply Utility Maximization?

The reverse direction is subtler.

- For $n = 2$ goods: yes, a choice function satisfying WARP can always be rationalized by some preference relation.
- For $n \geq 3$: **not necessarily**. WARP is not strong enough to guarantee the transitivity of the underlying revealed-preference relation, so a consistent utility may fail to exist.

(Exercise 2.9, Jehle & Reny.) Construct a three-good example with a choice function that satisfies WARP but cannot be rationalized by any utility function.

6. The Strong Axiom of Revealed Preference (SARP)

Axiom 6.1 — Strong Axiom (SARP)

For any finite sequence of distinct bundles $x^0, x^1, \dots, x^K$ such that

$$x^0 \succsim^R x^1, \quad x^1 \succsim^R x^2, \quad \dots, \quad x^{K-1} \succsim^R x^K,$$

we have $x^K \not\succ^R x^0$.

Here $\succ^R$ denotes “directly revealed preferred to.” SARP rules out cycles of *any* length, not just length 2 (which is WARP).

A choice function satisfies SARP if and only if it can be rationalized by some strictly monotone, strictly quasiconcave utility function. Equivalently:

- The Slutsky matrix $S(p, y)$ is **symmetric** as well as negative semidefinite.
- Demand is homogeneous of degree 0 in (p, y) .
- Budget balance holds: $p \cdot x(p, y) = y$.

7. Worked Examples

Example

(a) Take $p^0 = (1, 3)$, $x^0 = (4, 2)$ and $p^1 = (3, 5)$, $x^1 = (3, 1)$. Check WARP.

$$p^0 \cdot x^0 = 4 + 6 = 10, \quad p^0 \cdot x^1 = 3 + 3 = 6, \quad p^1 \cdot x^0 = 12 + 10 = 22, \quad p^1 \cdot x^1 = 9 + 5 = 14.$$

At prices p^0 : x^1 was affordable ($6 \leq 10$), so x^0 is revealed preferred to x^1 . At prices p^1 : x^0 costs $22 > 14 = y^1$, so x^0 was *not* affordable. **Consistent with WARP.** ✓

Example

(b) Take $p^0 = (1, 6)$, $x^0 = (10, 5)$ and $p^1 = (3, 5)$, $x^1 = (8, 4)$.

$$p^0 \cdot x^0 = 10 + 30 = 40, \quad p^0 \cdot x^1 = 8 + 24 = 32, \quad p^1 \cdot x^1 = 24 + 20 = 44, \quad p^1 \cdot x^0 = 30 + 25 = 55.$$

x^1 was affordable at p^0 ($32 \leq 40$), but x^0 was not affordable at p^1 ($55 > 44$). **Satisfies WARP.** ✓

Reference: Jehle & Reny, Advanced Microeconomic Theory, Chapter 2.